



SOI × Cosmosoc

Geo Snap



Land-Use Classification & Explainability from Space

Data Science Problem Statement

Event Details

Duration : 2 weeks

Team Size: 1 to 4 members

Submission : [Git Hub Repo Link](#)

Introduction

Satellite imagery forms the backbone of modern geospatial intelligence. From monitoring crop health to planning infrastructure, from protecting forests to disaster response - AI has become essential in understanding our planet.

Geo Snap challenges you to build intelligent models that can classify land-use from satellite imagery and explain why those predictions are made. You will work with real Sentinel-2 data from the EuroSAT dataset across two imaging modalities:

- RGB images (3 bands) - what a camera sees
- Multispectral Sentinel-2 images (13 bands) - what the human eye cannot see

Each image is a 64×64 pixel patch belonging to one of 10 land-use classes:

#	Class	#	Class
0	AnnualCrop	5	Highway
1	Forest	6	Industrial
2	HerbaceousVegetation	7	PermanentCrop
3	Pasture	8	Residential
4	River	9	SeaLake

Sentinel-2 Bands Reference

Understanding the spectral bands is key to unlocking multispectral modeling.

Resolution	Band	Description
10 m	B2	Blue
	B3	Green
	B4	Red
	B8	Near-Infrared (NIR)
20 m	B5, B6, B7	Vegetation Red Edge
	B8A	Narrow NIR
	B11, B12	Shortwave Infrared (SWIR)
60 m	B1	Coastal Aerosol
	B9	Water Vapor
	B10	Cirrus

Dataset Organization

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The dataset is provided in two modalities:

 EuroSAT/ ← RGB images (.jpg)

```
|—— train/
| |—— AnnualCrop/
| |—— Forest/
| |—— ... (10 class folders)
|—— val/
|—— ... (10 class folders)
```

 EuroSATallBands/ ← 13-band Multispectral images (.tif)

```
|—— train/
| |—— ... (10 class folders)
|—— val/
|—— ... (10 class folders)
```

 train.csv ← Official training split

 validation.csv ← Official validation split

 label_map.json ← Class index mapping

 EuroSAT_test_flat/ ← Flattened RGB test images (unlabeled)

 EuroSATallBands_test_flat/ ← Flattened Multispectral test images (unlabeled)

All images are 64×64 patches. The dataset link will be shared separately.

Task 1 - Land-Use Classification (RGB & Multispectral)

1A. RGB Classification

Input	64×64×3 RGB image
Output	10-class probability distribution + predicted label
Models Allowed	Any CNN, Transformer, or ML model
Goal	Baseline land-use classification using visual data only

1B. Multispectral Classification

Input	64×64×13 Sentinel-2 image (all 13 bands)
Output	10-class probability distribution + predicted label
Approach	Direct multi-band deep learning or constructed spectral features (e.g., NDVI, NDWI, SWIR indices)
Goal	Investigate whether spectral information improves classification over RGB alone

Objective

Investigate whether spectral information improves land-use discrimination – especially for classes that are visually similar in RGB but spectrally distinct (e.g., AnnualCrop vs. PermanentCrop, or Pasture vs. HerbaceousVegetation).

What We Encourage

- Explore robust feature extraction from RGB data
- Handle inter-class visual ambiguity
- Analyze class-wise performance and failure modes
- Compare RGB-only vs. multispectral approaches
- Study the role of spectral diversity in classification performance

Task 2 - Explainability & Model Interpretation (XAI)

In real-world geospatial systems, model accuracy alone is not sufficient. Understanding why a model makes a decision is critical for trust, deployment, and scientific insight – especially in agriculture monitoring, disaster management, environmental protection, and urban planning. Participants must provide interpretable explanations for the predictions made by their models from Task 1.

2A. Visual Explainability

- Generate heatmaps or saliency maps for RGB and/or multispectral models
- Show which spatial regions the model considers important
- Include examples for both correct and incorrect predictions

2B. Spectral / Feature Explainability

- Analyze band importance (which of the 13 bands matter most?)
- If spectral indices (NDVI, NDWI, etc.) were used, explain their role
- Connect your interpretations to established remote sensing principles

2C. Error & Confusion Analysis

- Present a confusion matrix for your model(s)
- Identify and explain major misclassification patterns
- Provide spectral and/or visual reasoning for why certain classes are confused

Task 3 – Bonus: Environmental Insights

(Open-Ended Creative Task – Geospatial Reasoning & Domain Insight)

This bonus task rewards creativity, scientific reasoning, and real-world geospatial insight rather than raw model accuracy.

Explore the multispectral data and/or your classification outputs to extract meaningful environmental, ecological, or urban insights. There is no single correct approach.

Example Directions (not exhaustive)

- Vegetation & crop health patterns – NDVI analysis across agricultural classes
- Water & moisture analysis – NDWI, SWIR-based moisture mapping
- Urban/built-up land behavior – spectral signatures of residential vs. industrial areas
- Land suitability & geo-planning – insights for land-use planning

Participants are strongly encouraged to explore beyond the topics listed above. Surprise us.

Deliverables & Submission Format

All submissions must be made as a single GitHub repository containing code, results, and documentation.

Required Deliverables

Deliverable	Details
Predictions (Task 1)	rgb_predictions.csv → img_id, predicted_label
	ms_predictions.csv → img_id, predicted_label
Code (All Tasks)	Full training & inference code, well-commented notebooks

Deliverable	Details
Explainability (Task 2)	Visual explanations, band/feature analysis, error & confusion analysis – in a separate Jupyter notebook and discussed in the report
Technical Report	Max 10 pages covering methodology, results, and analysis
README.md	Setup instructions, dependencies, and how to reproduce results

Submission Rules

- 1.Upload all files to one GitHub repository named: TeamName_Aurora_submission
- 2.Submit only the GitHub repo link before the deadline
- 3.Late submissions will not be accepted

Scoring & Evaluation

Stage 1 - Online Hackathon (60% of Total Score)

Component	Weightage
Task 1 - Land-Use Classification	30%
Task 2 - Explainability & Model Interpretation	30%
Task 3 - Bonus Environmental Insights	15%
Report + README Quality & Reproducibility	25%
Total (Stage 1)	60%

Only teams shortlisted based on Stage 1 performance will proceed to Stage 2.

Only teams with submissions following the instructions properly are considered.

Stage 2 - Final Presentations (40% of Total Score)

Shortlisted teams only.

- presentation before the panel
- Evaluation based on:
 - Technical depth
 - Clarity of explanation
 - Innovation and creativity
 - Justification of modeling choices
- Includes a Q&A session with the panel

Presentation Guidelines

- Teams must submit their presentation slides (.ppt or .pdf) before the official deadline for PPT submission.
- The presentation time is strictly 10 minutes, followed by 5 minutes for the Q&A session with the panel.

Contact & Important Notes

- Only the team leader of each team must join the WhatsApp group (link shared after registration)

Event Lead

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